

## **PROBLEM**

### **1. Decision Tree Classification – Problem Statement**

Develop a Decision Tree Classification system that can classify data into predefined categories based on a set of input features. The dataset should first be loaded and examined for its structure, data types, missing values, and other preprocessing requirements. The relevant features and target variable must then be identified and the dataset divided into suitable training and testing sets.

Construct the Decision Tree using an appropriate splitting criterion such as Gini Index or Information Gain. The model should learn decision rules from the training data and use these rules to predict the classes of unseen test instances. The resulting tree structure should be visualized to understand how the features contribute to classification.

Evaluate the trained model using accuracy, precision, recall, F1-score, confusion matrix, and classification report. Identify misclassified instances and analyze possible reasons for incorrect predictions. The experiment should demonstrate how Decision Tree algorithms can be used for interpretable and effective supervised classification.

### **2. Q-Learning – 4×4 Grid World Problem Statement**

Develop a Q-Learning reinforcement learning system for navigating an agent through a 4×4 Grid World. The environment should contain a starting state, a goal state, and at least one obstacle. The agent should be allowed to perform four actions: Up, Down, Left, and Right.

Design an appropriate reward system in which the agent receives a positive reward for reaching the goal and penalties for hitting obstacles or making unnecessary movements. Initialize a Q-table for storing the expected rewards associated with each state-action pair. The agent should use an  $\epsilon$ -greedy strategy to balance exploration of new actions with exploitation of previously learned actions.

Train the agent for at least 100 episodes and update the Q-table after every interaction using the Q-Learning update equation. Record and compare the Q-values at Episodes 1, 50, and 100 to observe how the agent's knowledge changes during training.

After training, determine the best action for each state based on the highest Q-value and display the learned policy